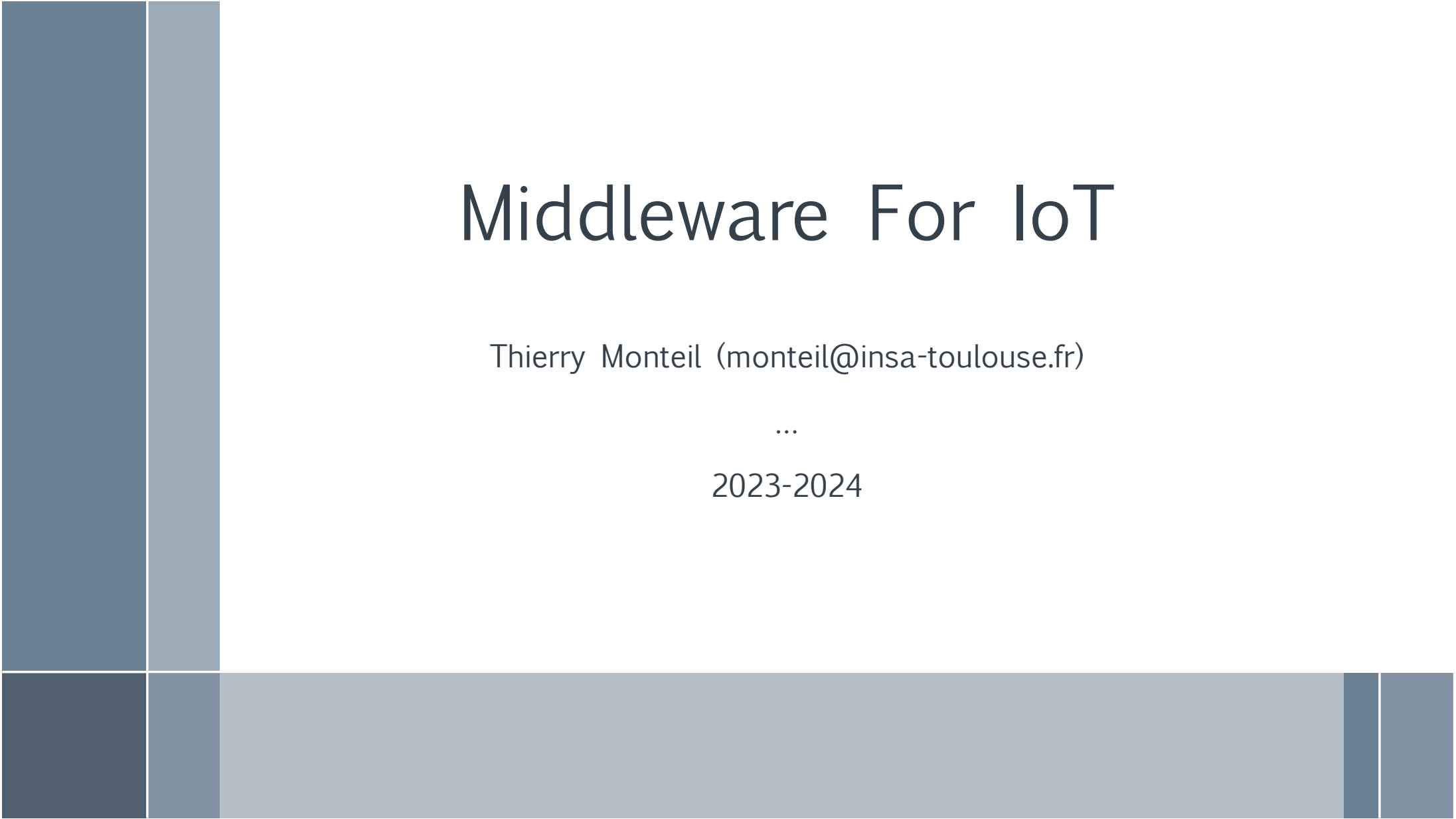




Middleware For IoT

Thierry Monteil (monteil@insa-toulouse.fr)

...

2023-2024



Organisation



- › Mix of MOOC and practice
- › Planning
 - 1/2h-1h15 presentation
 - 2*1,25h MOOC alone (classroom or where you like)
 - 2*2,75h Practical work: simple IoT architecture with MQTT
 - 2,75h Practical work: oneM2Mstandard – standardized architecture and service for IoT
 - 2,75h Practical work: user friendly development – nodeRED
 - Evaluation: report
 - One week after the last practical work
 - To be included in the portfolio also
- › French version of the MOOC (with semantic part):
 - <https://openclassrooms.com>
 - Mettez en place une architecture pour objets connectés avec le standard oneM2M
- › English version of the MOOC (without semantic part):
 - <https://open.insa-toulouse.fr> => MOOC
 - Internet of Things with oneM2M standard
- › On moodle Machine to machine (Key: tarifa) : <https://moodle.insa-toulouse.fr/course/view.php?id=785>

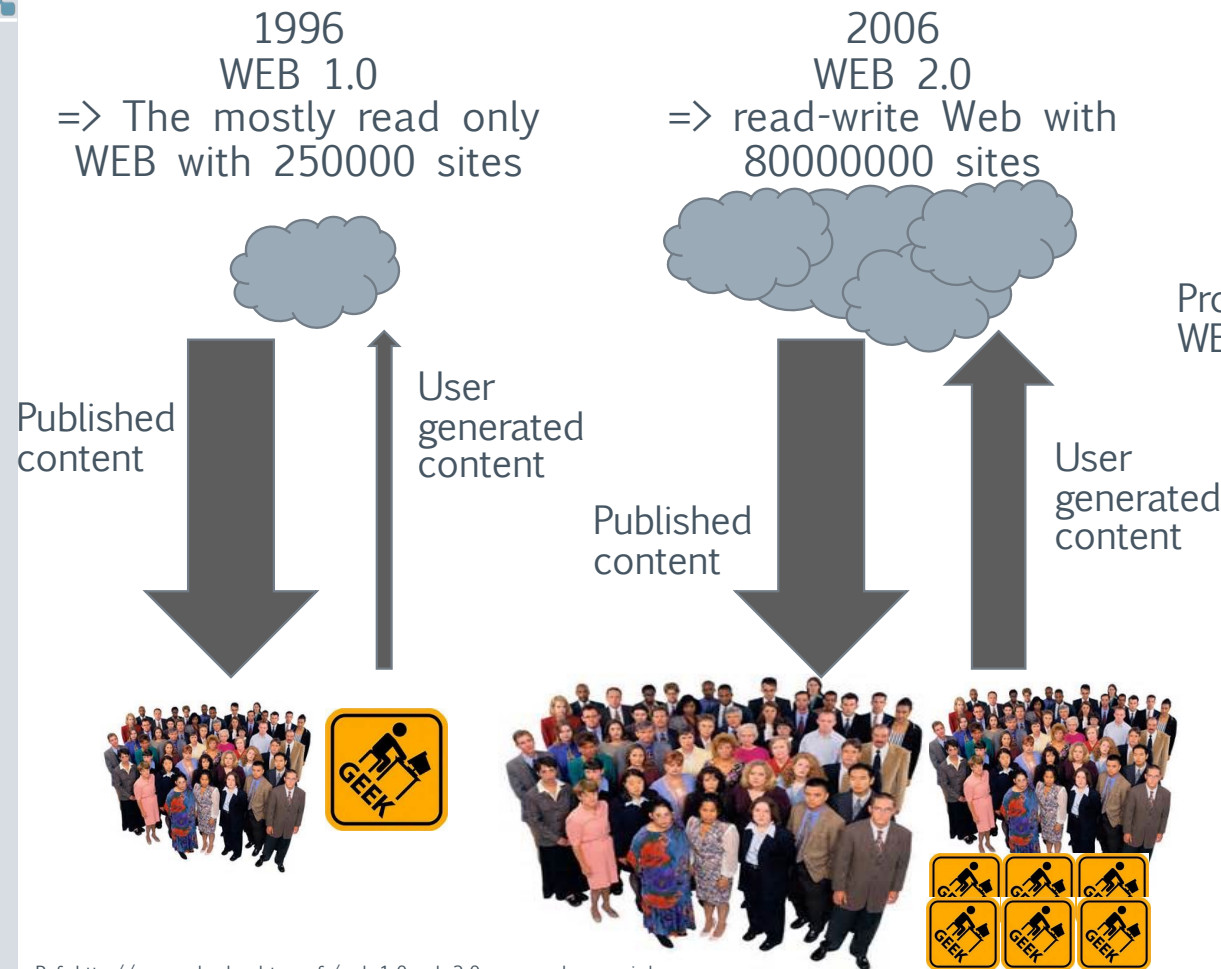


Good practice

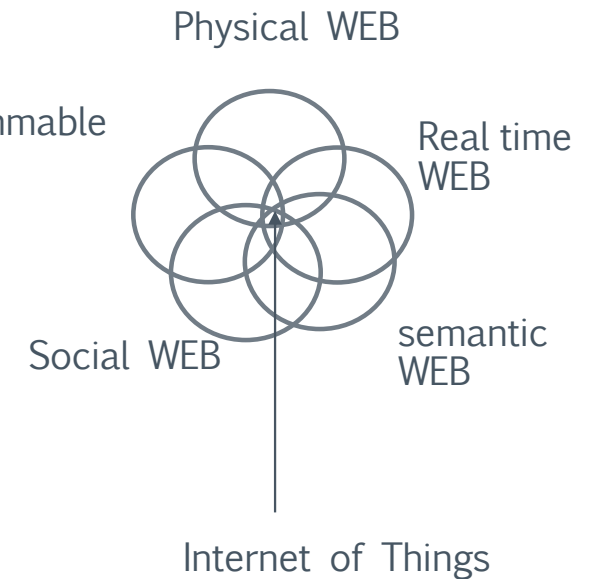


- › Start the MOOC now!
- › Before the first practical work with teacher :
 - Finish the MOOC video
- › Before practical work install all the software needed:
 - IDE Arduino+lib for ESP8266, node-RED, ACME oneM2M, MQTT, etc
- › Come in practical work with your laptop
- › !! Connection with SOA course

Evolution of WEB



Few years later ...
WEB 3.0 / Web of things²



Ref: <http://www.webschool-tours.fr/web-1-0-web-2-0-vous-voulez-savoir-la-difference/>

2-<http://www.webofthings.org/2011/02/04/lift11-talk-transcript/>

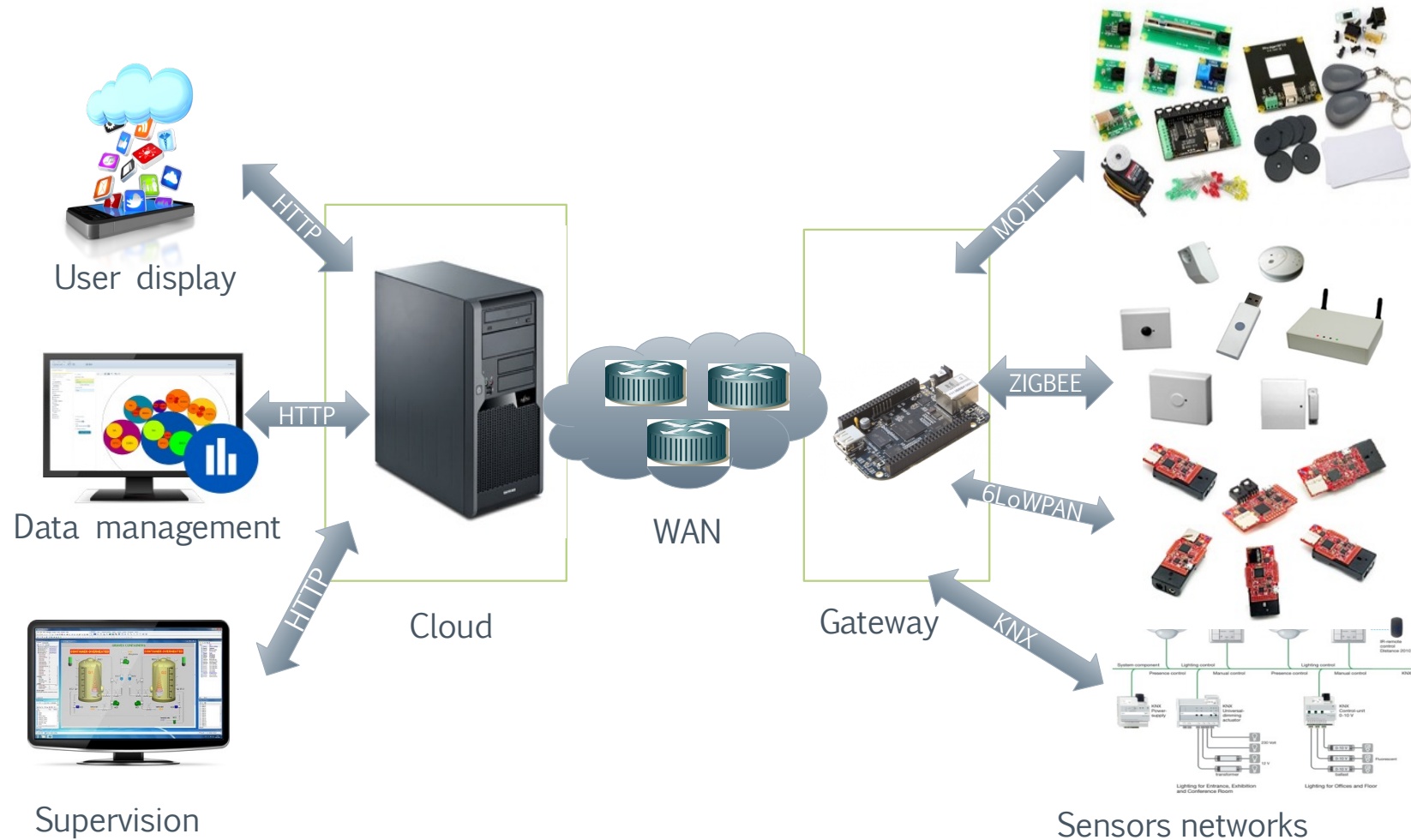
Application domains



M2M World of Connected Services The Internet of Things

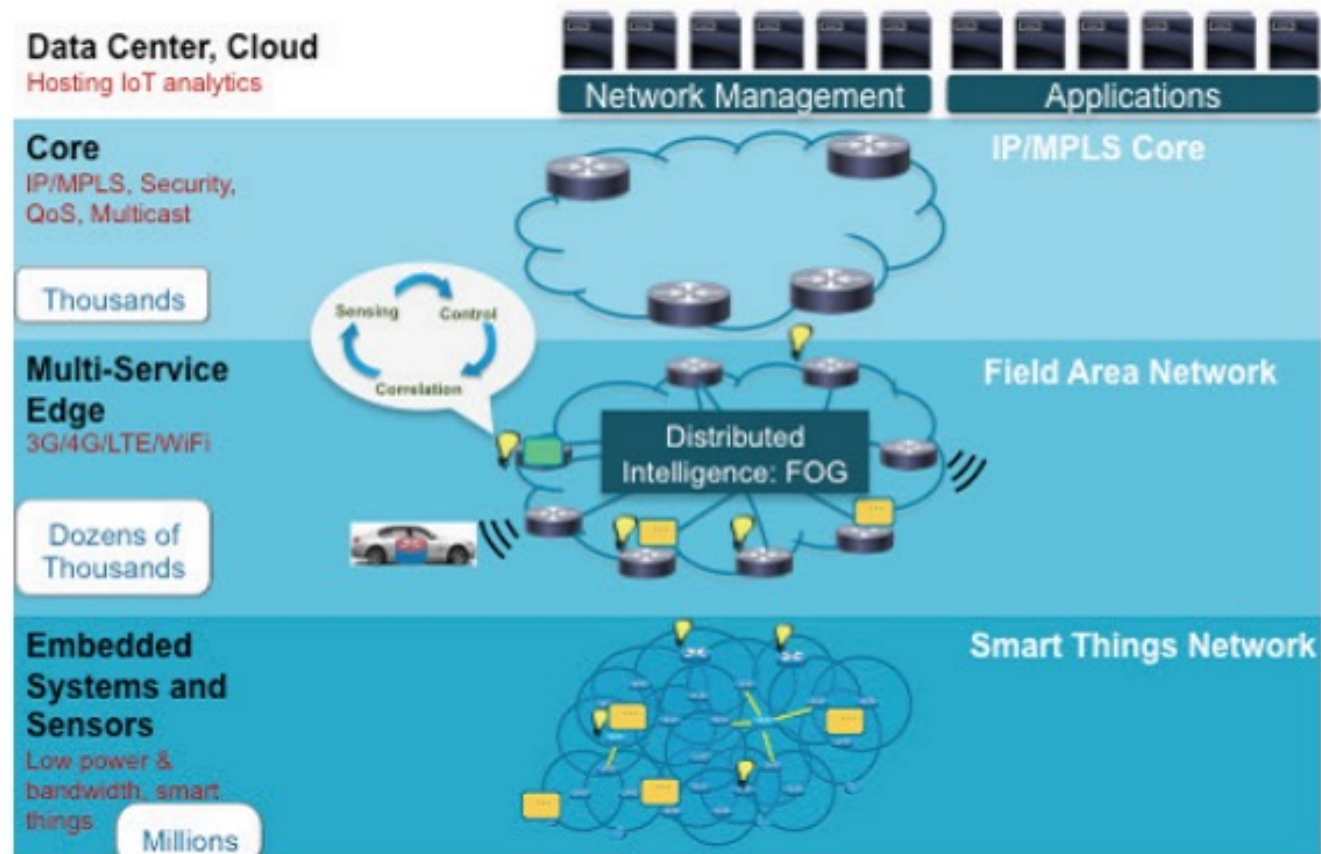


Classical architecture



Fog architecture

The Internet of Thing Architecture and Fog Computing



market fragmentation



- The current marketplace is extremely fragmented
 - Increase the R&D cost in each specific domain.
 - Silo model is not an efficient way to communicate, it is a barrier to further development.
 - Many vertical solutions have been designed independently and separately for different applications, which impedes large-scale interoperable deployment.



IoT Deployment “Pain Points”



→ Different IoT sub-systems use different underlying technologies

IoT Underlying Connectivity Networks

E.g. Cellular, Fixed, Wi-Fi, ...

IoT Security Models

E.g. DTLS, TLS, PSK, PKI, ...

IoT Transport Protocols

E.g. HTTP(s), CoAP(s), MQTT(s), WebSockets, ...

IoT Content Serializations

E.g. XML, JSON, CBOR, Plain-Text, ...

IoT Services and Management Technologies

E.g. OCF, LWM2M, Thread, OMA DM, TR-069, ...

IoT Semantic Ontologies

E.g. SAREF, W3C TD, ...

→ Building, deploying and maintaining a system with diverse combinations of technologies can be challenging

Manufacturing/ Industry Automation

Healthcare

Cities

Farming/
Agrifood



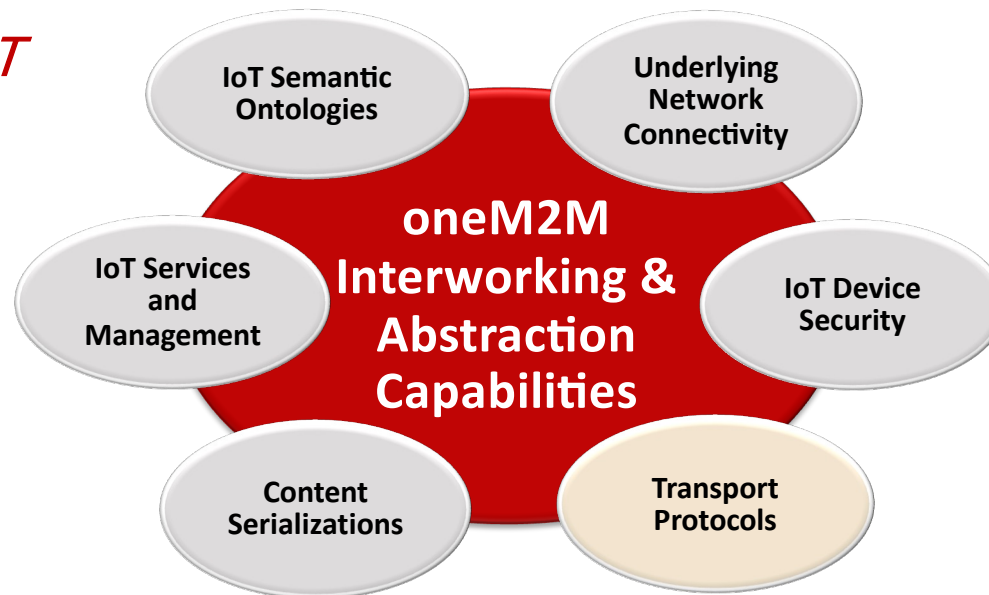
Source: AIOTI WG3 (IoT Standardisation) – Release 1.2



oneM2M's Value Proposition



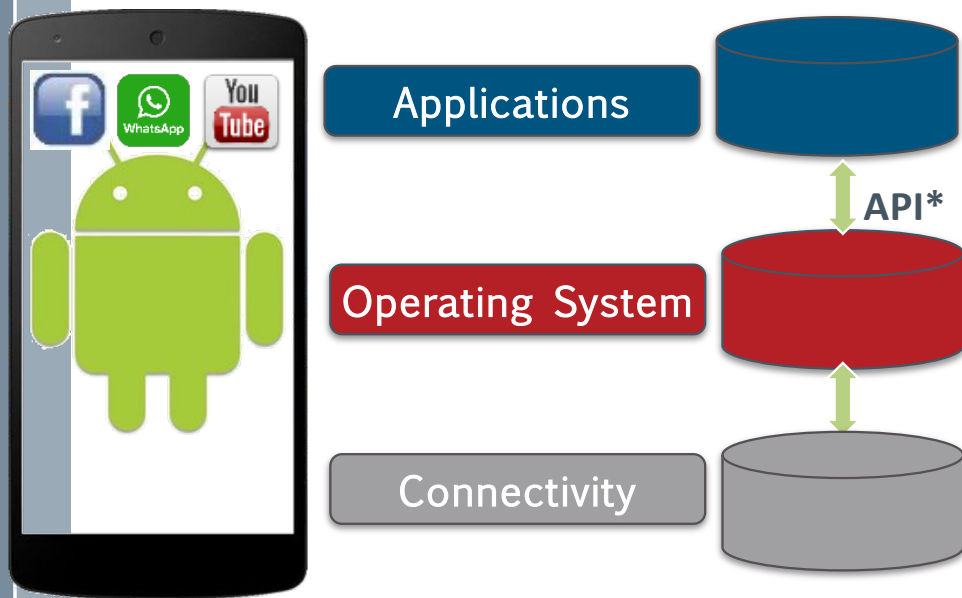
Interwork different underlying IoT technologies, devices, applications and data together with one another



Abstract and hide the complexity of these different forms of interworking from the various stakeholders

Simplify life for IoT Stakeholders!!!

Take a lesson from the modularity of smart phones that offer a framework for developers!



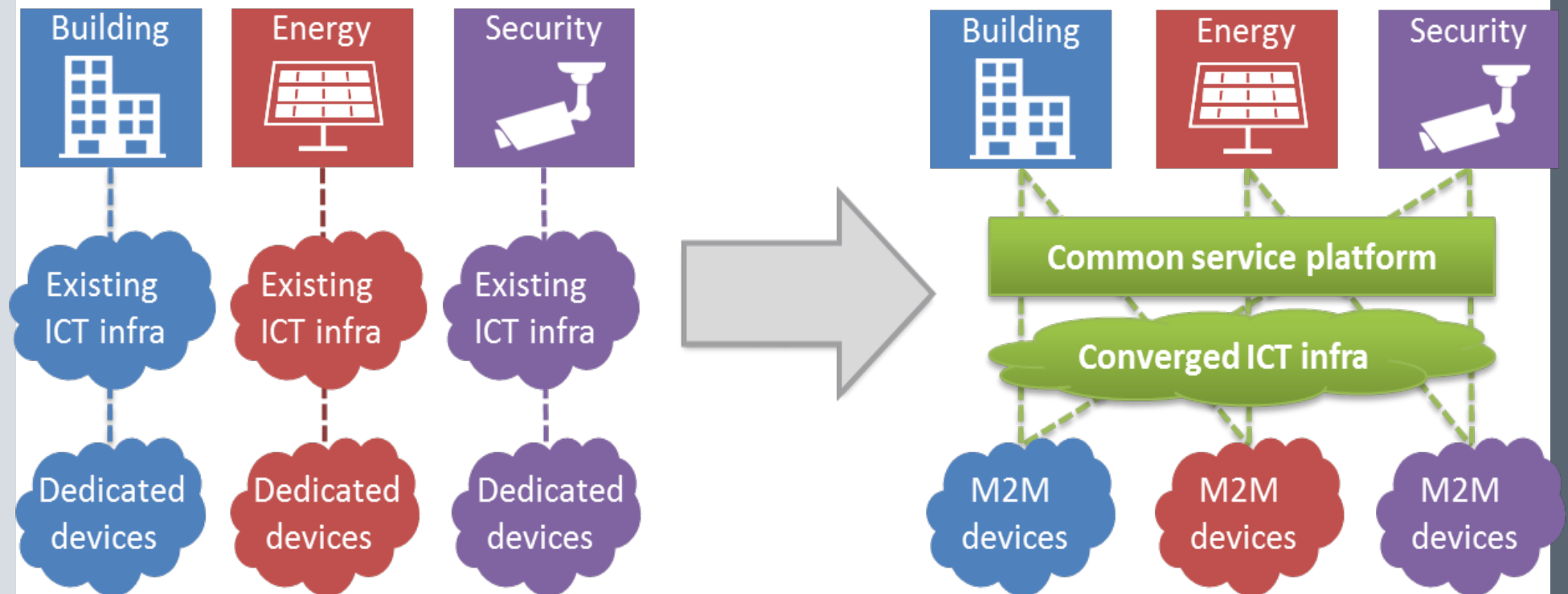
* API: Application Programmers Interface

- **Applications** access the Connectivity Layer and built-in sensors, via API's provided by the Operating System
- **Operating System** collects connectivity requests from applications, buffers messages, optimizes & controls device's network use, etc.
- **Connectivity Layer** provides device access to the Internet via the 3GPP mobile network, Wi-Fi, etc.



Apply similar concepts to
IoT!

oneM2M Breaks Down the Silos



Standardized Architecture of OneM2M



Reference Point

Common Services Entity

Application Entity

Network Services Entity

Node
device

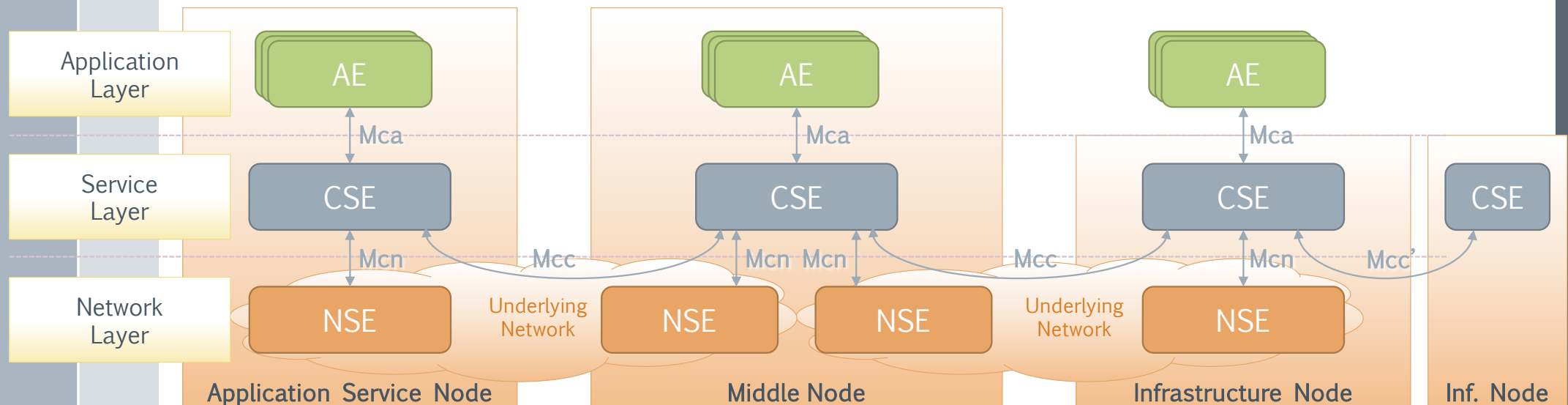
One or more interfaces - Mca, Mcn, Mcc and Mcc' (between 2 service providers)

Provides the set of "service functions" that are common to the M2M environments

Provides application logic for the end-to-end M2M solutions

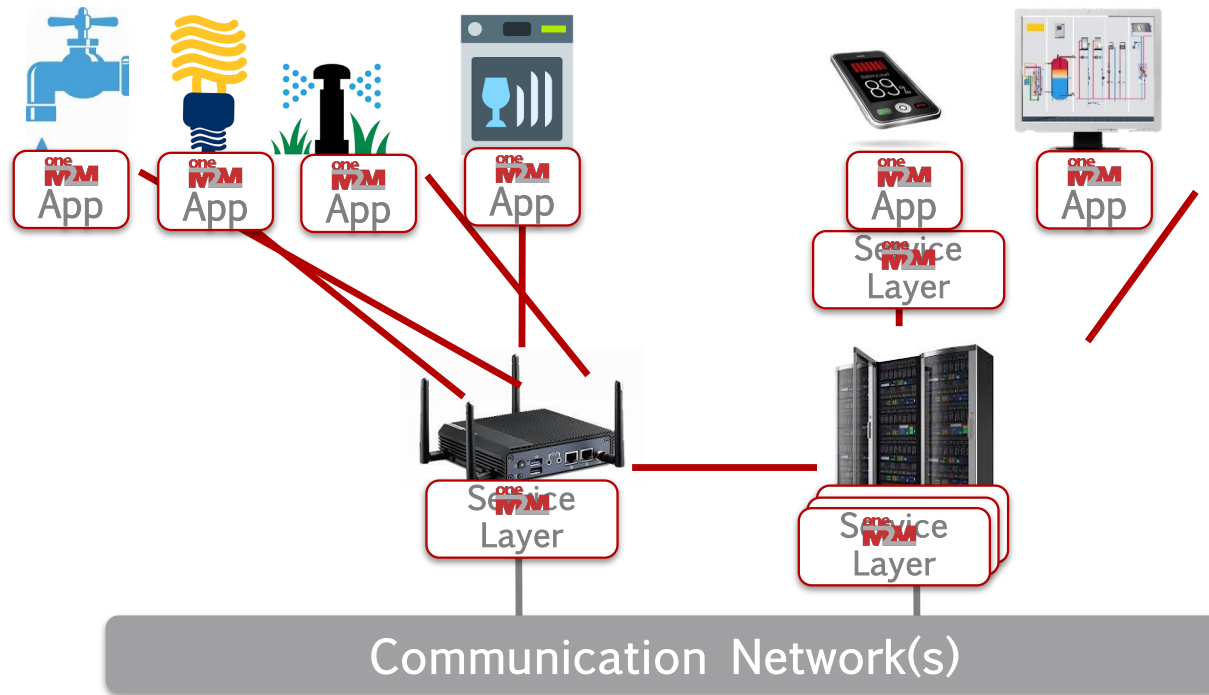
Provides services to the CSEs besides the pure data transport

Logical equivalent of a physical (or possibly virtualized, especially on the server side)



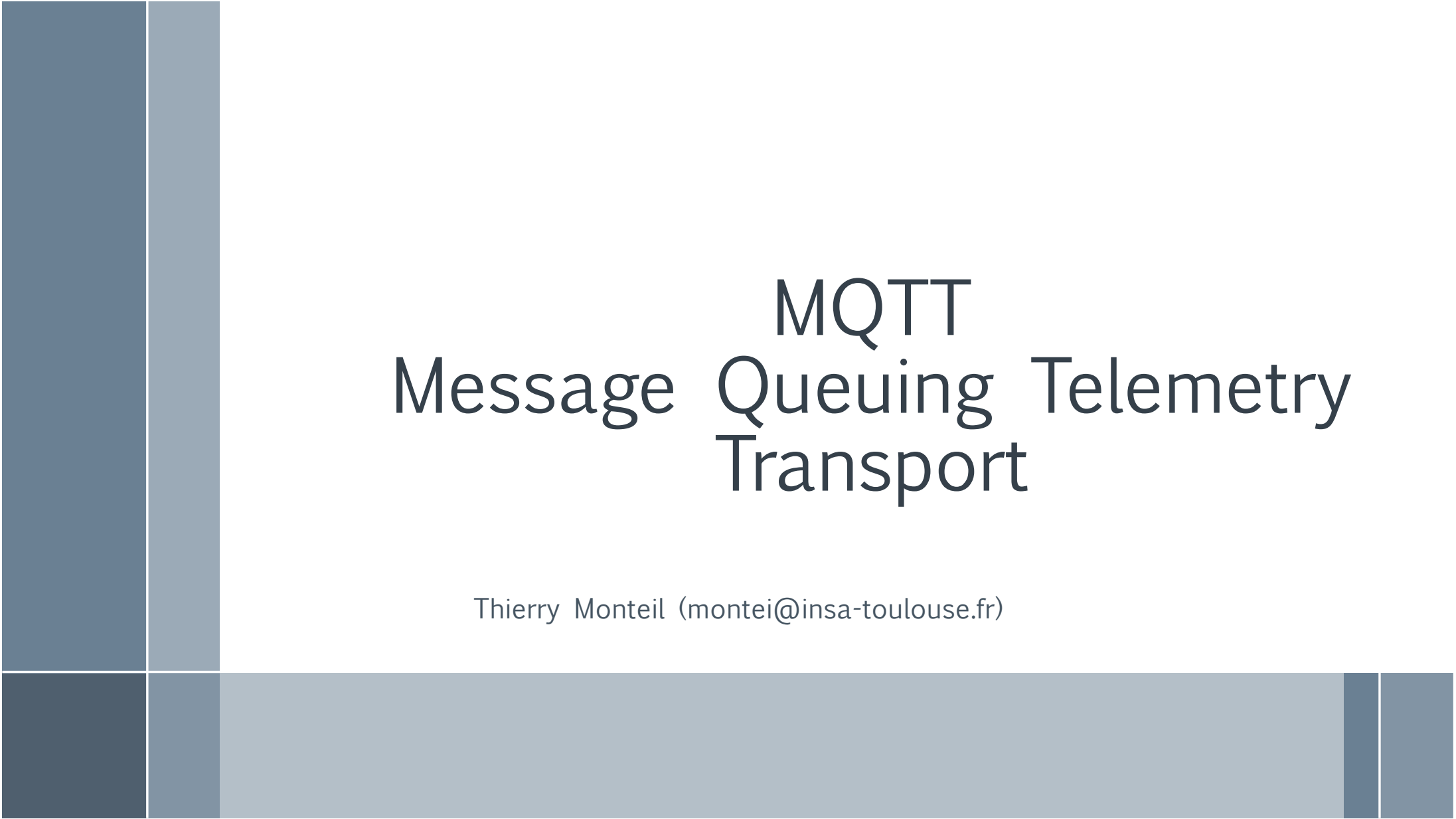
From oneM2M Service Layer Platform – Initial Release: Omar Elloumi/Nicolas Damour

oneM2M is Distributive and Scalable



Flexible Deployment Options

- IoT Cloud / Enterprise
- IoT Gateway
- IoT Edge Device
- IoT User Devices



MQTT

Message Queuing Telemetry Transport

Thierry Monteil (montei@insa-toulouse.fr)

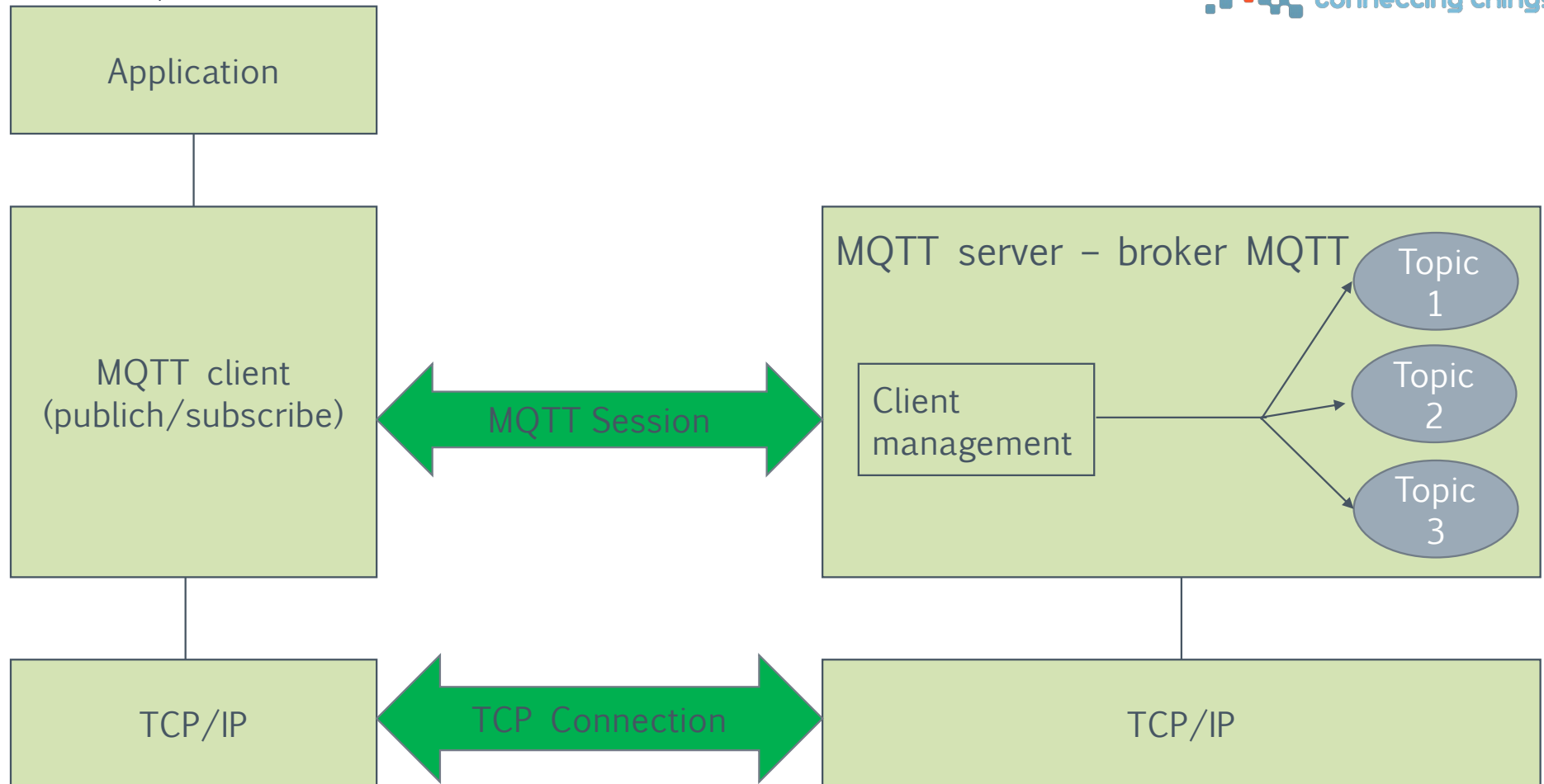


MQTT



- › Created by IBM and Eurotech
- › Several version 3.1.1, 5
- › Lightweight message queuing
- › Made first for sensor and actuator
- › Key points:
 - Asynchronous communication : Publish / subscribe model
 - Low overhead (2 bytes header)
 - Several messages queues based on topics concept
 - Simple protocol: low complexity, low power
 - Based on TCP protocol
 - Secure communication could be added

MQTT architecture



Node-RED Fast prototyping

Samir MEDJIAH – medjiah@laas.fr (Associate Professor at INSA – Univ. of Toulouse / Researcher at LAAS-CNRS)

Thierry Monteil – monteil@insa-toulouse.fr (Professor at INSA – Univ. of Toulouse / Researcher at IRIT)



Node-RED



- › Created by IBM as opensource
- › Flow-based programming
- › Applications are based on set of blocks with: inputs, outputs and functionalities
- › Use of browser to create and deploy flows on internet
- › Based on javascript and node.js and store in json files
- › Simple to install and creat web access to application
- › Large community and API (<https://flows.nodered.org>)

Node-Red IDE

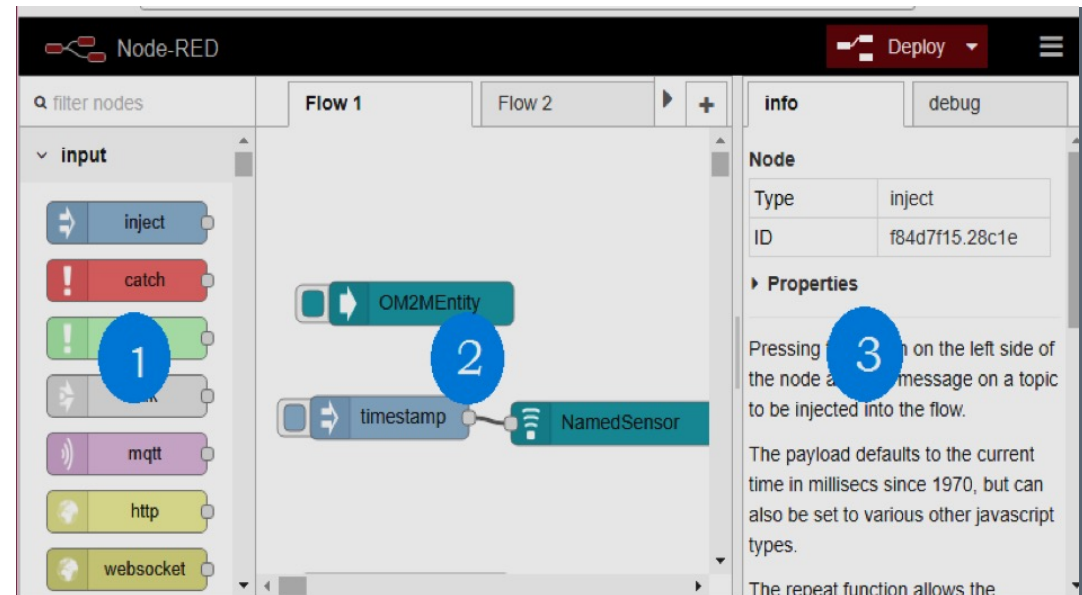
- Three panels:

1. Node Panel: list of nodes

2. Flow Panel:

- workspace of the user,
- Creation of streams by dragging and dropping

3. Information and Debug Panel:





Conclusion



- › IoT is everywhere
 - › Many solutions exists to create IoT systems
 - › Links with hardware, network, internet, cloud, security, Data management, IA, ...
 - › Difficulties to manage: different technologies, interoperability, security, quality of service
 - › Easy to create a prototype more difficult to develop a product
- => Easy to solve concrete problem
- => Fun application